

Work and energy

Kinetic energy, potential energy and work

YOUR AIM TODAY

Track energy transfers and use work-energy equations.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

The ideas you need

Energy is measured in joules; work is energy transferred by a force through a distance.

- Work by constant force along motion: $W=Fs$.
- Kinetic energy $KE=(1/2)mv^2$.
- Gravitational potential energy $GPE=mgh$.
- Work done by the resultant force equals change in KE.
- Against resistance, input work becomes KE, GPE and energy lost.

MEMORY RULE

Write an energy before-and-after line before inserting numbers.

Speed change

WORKED STEPS

2 kg from 3 to 7 m/s: change $KE=0.5(2)(49-9)=40$ J.

Lift

WORKED STEPS

5 kg raised 2 m gains $GPE=5g(2)=10g$ J.

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Find KE of 4 kg moving at 6 m/s.	
2. Find GPE gained by 3 kg raised 5 m.	
3. A 20 N force moves 8 m along its direction. Find work.	
4. A 2 kg body starts from rest; 36 J resultant work is done. Find speed.	
5. A 5 kg body falls 4 m from rest without resistance. Find speed.	
6. A 50 N force pulls 3 m while 10 N resistance acts. Find net work.	

STOP

Do not continue until all six questions have an attempt.

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	$0.5(4)(36)=72 \text{ J.}$
2	$3g(5)=147 \text{ J.}$
3	$20(8)=160 \text{ J.}$
4	$36=0.5(2)v^2$, so $v=6 \text{ m/s.}$
5	$mgh=0.5mv^2$, so $v=\sqrt{2gh}=8.85 \text{ m/s.}$
6	$(50-10)(3)=120 \text{ J.}$

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.